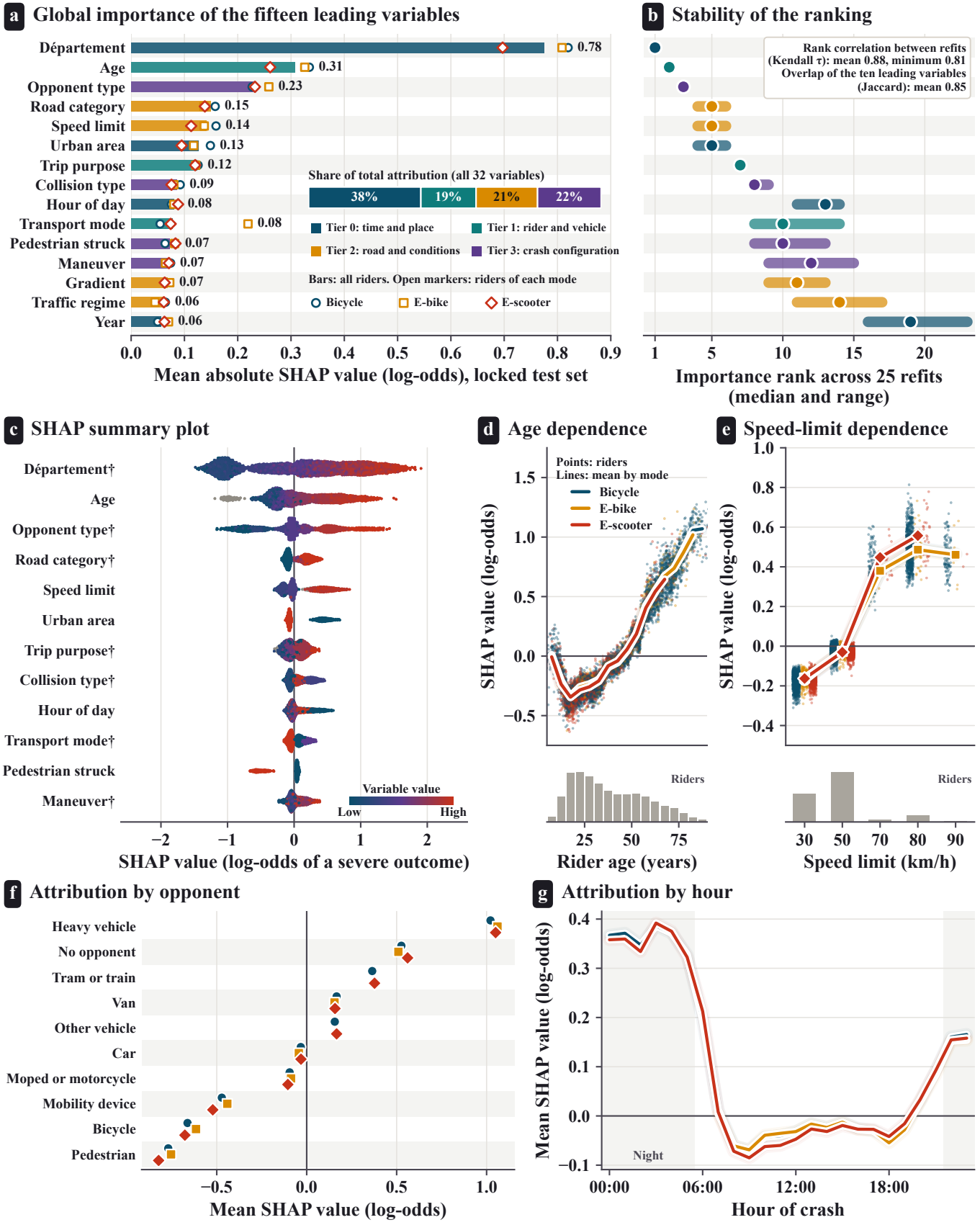

a
Global importance of the fifteen leading variables
Département
Age
Opponent type
Road category
Speed limit
Urban area
Trip purpose
Collision type
Hour of day
Transport mode
Pedestrian struck
Maneuver
Gradient
Traffic regime
Year
0.78
0.31
0.23
0.15
0.14
0.13
0.12
0.09
0.08
0.08
0.07
0.07
0.07
0.06
0.06
Share of total attribution (all 32 variables)
38%
19%
21%
22%
Tier 0: time and place
Tier 1: rider and vehicle
Tier 2: road and conditions
Tier 3: crash configuration
Bars: all riders. Open markers: riders of each mode
Bicycle
E-bike
E-scooter
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Mean absolute SHAP value (log-odds), locked test set
b
Stability of the ranking
Rank correlation between refits
(Kendall τ): mean 0.88, minimum 0.81
Overlap of the ten leading variables
(Jaccard): mean 0.85
1 5 10 15 20
Importance rank across 25 refits
(median and range)
c
SHAP summary plot
Département†
Age
Opponent type†
Road category†
Speed limit
Urban area
Trip purpose†
Collision type†
Hour of day
Transport mode†
Pedestrian struck
Maneuver†
Variable value
Low
High
−2 −1 0 1 2
SHAP value (log-odds of a severe outcome)
d
Age dependence
Points: riders
Lines: mean by mode
Bicycle
E-bike
E-scooter
SHAP value (log-odds)
1.5 1.0 0.5 0.0 −0.5
Riders
25 50 75
Rider age (years)
e
Speed-limit dependence
SHAP value (log-odds)
1.0 0.8 0.6 0.4 0.2 0.0 −0.2 −0.4
Riders
30 50 70 80 90
Speed limit (km/h)
f
Attribution by opponent
Heavy vehicle
No opponent
Tram or train
Van
Other vehicle
Car
Moped or motorcycle
Mobility device
Bicycle
Pedestrian
−0.5 0.0 0.5 1.0
Mean SHAP value (log-odds)
g
Attribution by hour
Mean SHAP value (log-odds)
0.4 0.3 0.2 0.1 0.0 −0.1
Night
00:00 06:00 12:00 18:00
Hour of crash